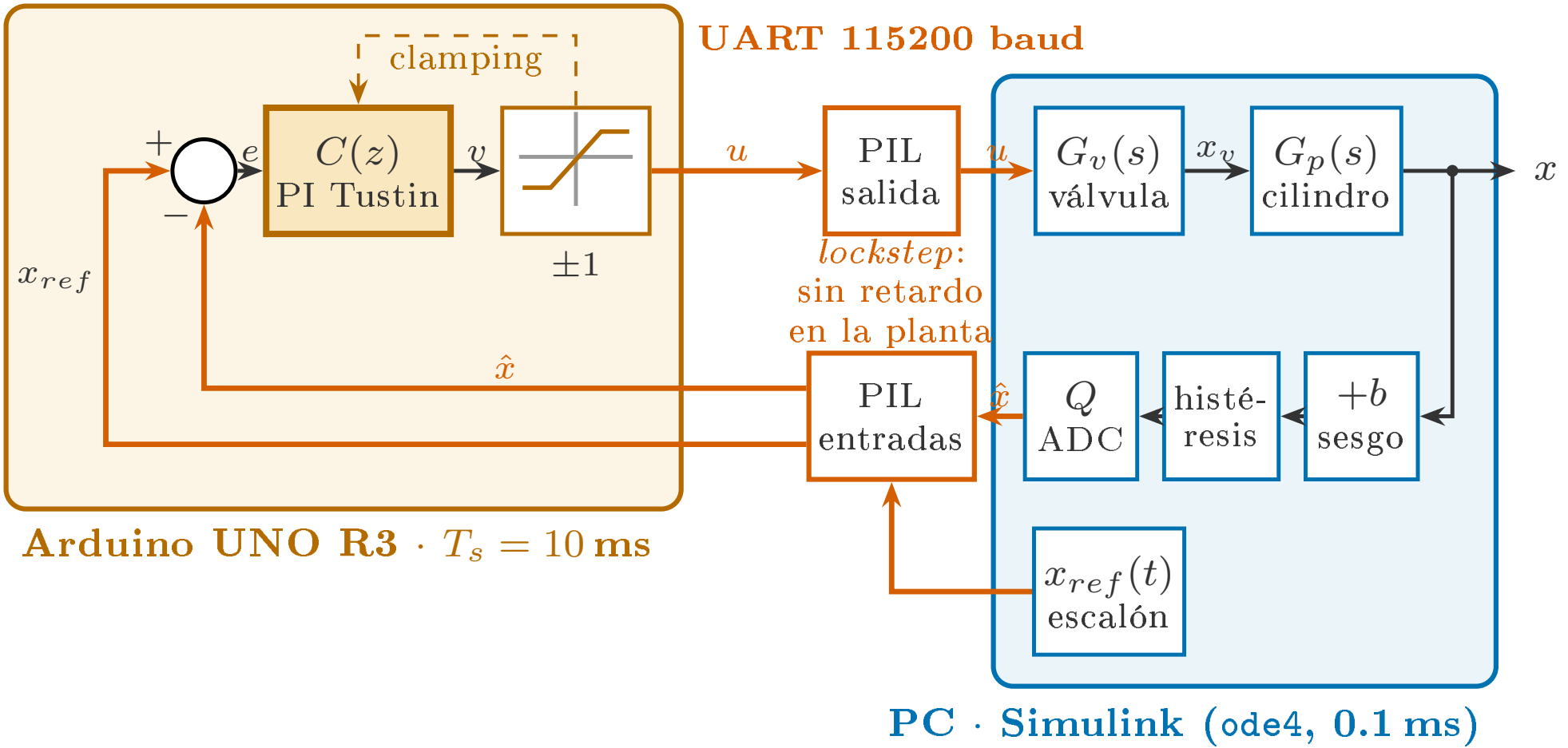

UART 115200 baud
clamping
+
−
e
C(z)
PI Tustin
v
±1
u
PIL
salida
u
$G_v(s)$
válvula
x_v
$G_p(s)$
cilindro
x
x_{ref}
lockstep:
sin retardo
en la planta
$\hat{x}$
PIL
entradas
$\hat{x}$
Q
ADC
histé-
resis
+b
sesgo
$x_{ref}(t)$
escalón
Arduino UNO R3 · $T_s = 10$ ms
PC · Simulink (ode4, 0.1 ms)